Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 5 · SLO 2 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 5 : SLO â€“ 2 | Matrix Multiplication

I. Fill in the Blanks

1. columns â€” valid only if columns of the first matrix equal rows of the second.
2. rows of first â€” the result has dimensions (rows of first) Ã— (columns of second).
3. corresponding â€” multiply each row element with the corresponding column element.
4. each element (repeated/dot-product) â€” a loop is required for each computation.
5. nested â€” use three nested loops (i, j, k).
6. zero â€” initialize result elements to 0.
7. dot â€” the innermost loop calculates the dot product.
8. accumulated (running sum) â€” sum += a[i][k] * b[k][j].
9. result â€” store the computed value in the result matrix res[i][j].
10. dimension mismatch (runtime/logical) error.
11. 2D arrays (contiguous memory) â€” matrix elements are stored in two-dimensional arrays.
12. use (computation) â€” initialize arrays before computing.
13. dimensions â€” declare the result with proper r1 Ã— c2 dimensions.
14. 2Ã—4 â€” (2Ã—3) Ã— (3Ã—4) â†’ 2Ã—4.
15. integer (element-wise multiply-accumulate) arithmetic.